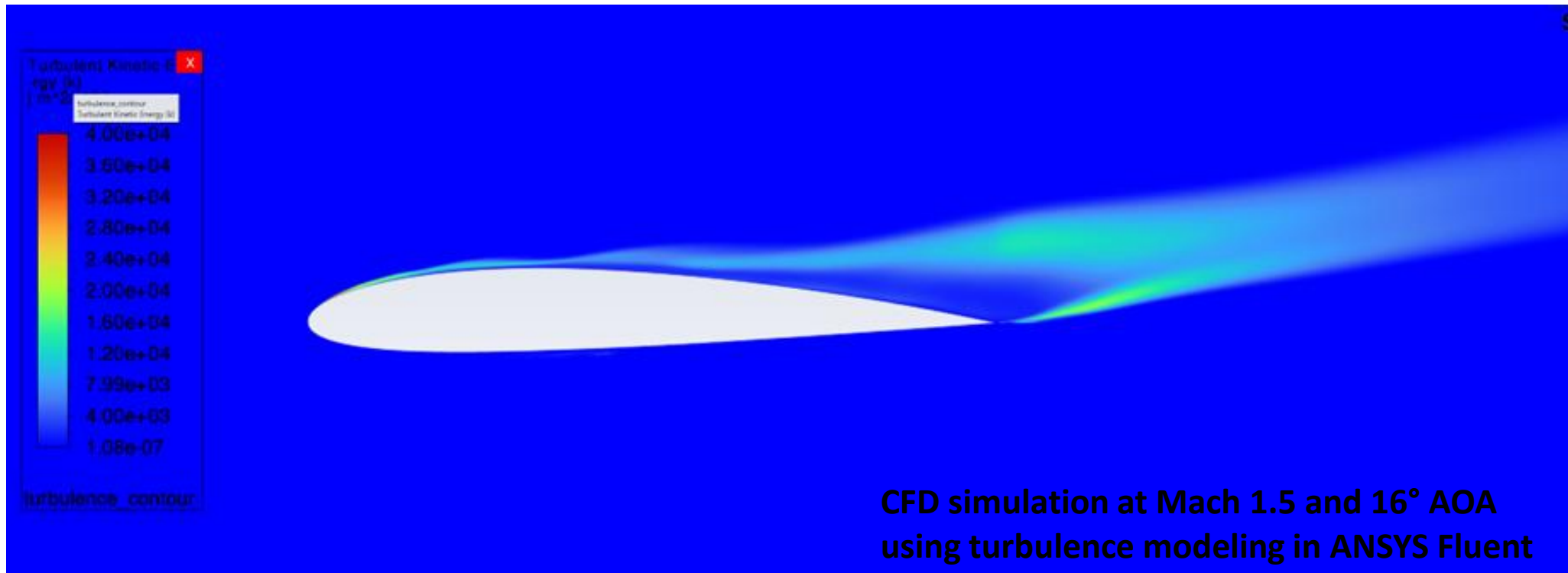


Flow over a 2-D 2412 Airfoil

Gerardo Sandoval



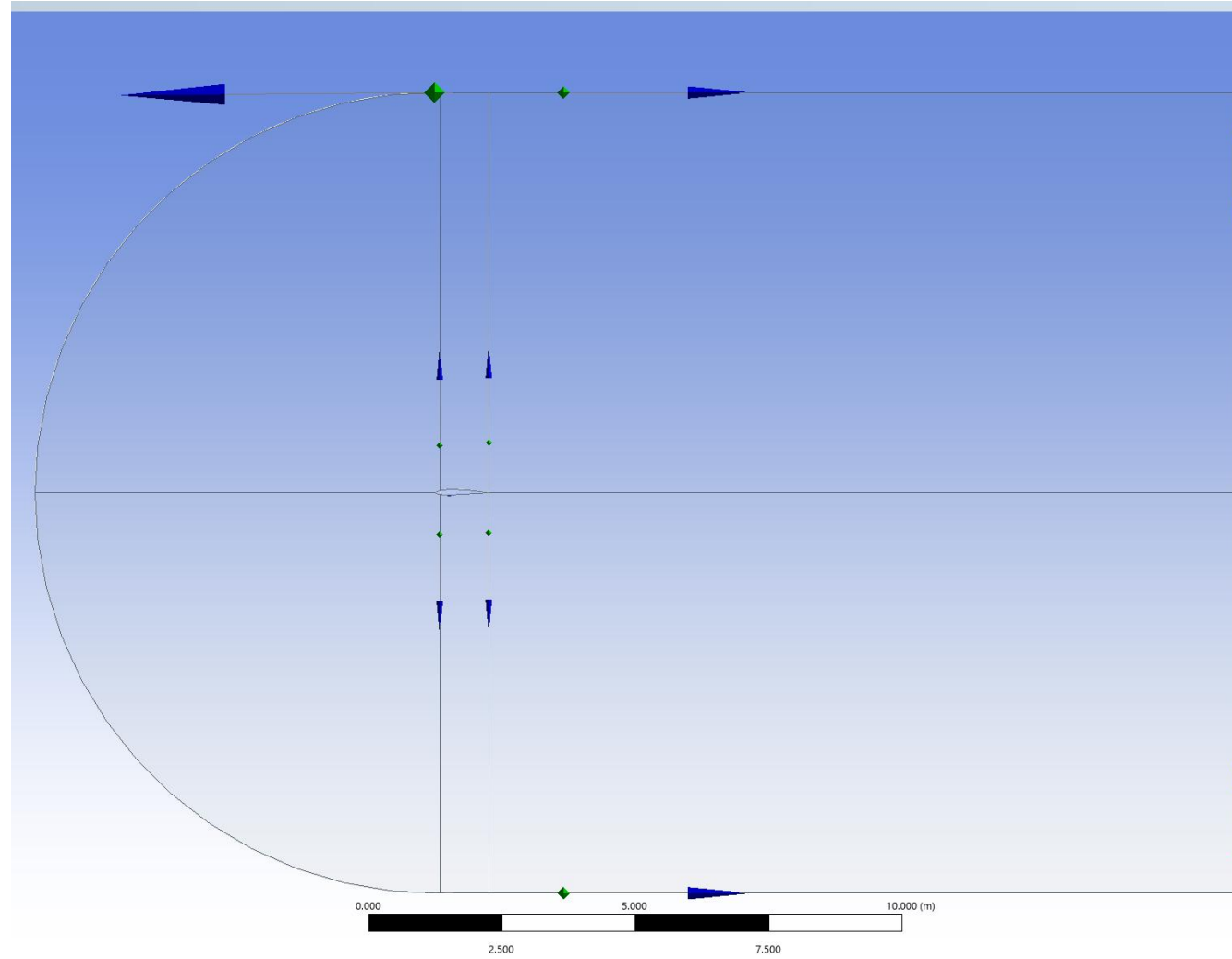
Overview of Presentation

- Project Background
- Methods
- Figures
- Validation and Verification
- Key Results and Conclusion
- References and Appendix

Project Background

- Goal – Lift and Drag Coefficients and Flow Contours
- 2-D Flow over Airfoil is simple.
- Mach speeds of 0.13, 1.0, and 1.5.
- Ansys Fluent Results are compared to NASA experimental data.
- At Mach 1.0 and 1.5, density-based solver is used to capture turbulence.

Figure 1. Geometry of the Airfoil and the Fluid Domain in Ansys Design Modeler.



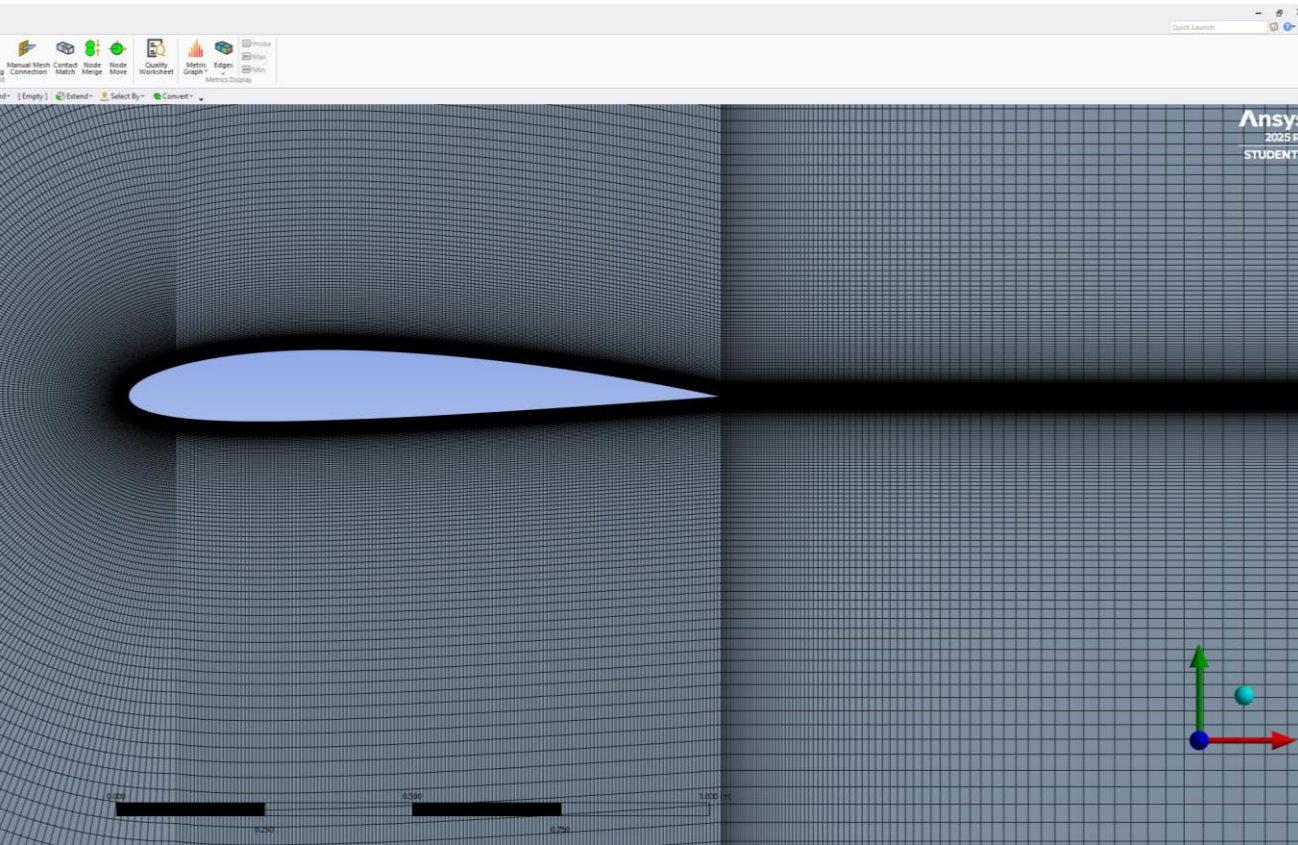


Figure 2. Structured Mesh

Methods - Ansys Workbench

- Created 2-D Airfoil Geometry with Excel and coordinate data.
- Generated Mesh and Defined Boundary Conditions
- Ran Fluent using
 - 2-D Flow
 - Viscous -> SST k-omega , 1000 Iterations.
 - Mach 0.13 at 0-, 8-, and 16-degrees angle of attack.
 - Mach 1.0 at 0- degree angle of attack.
 - Mach 1.5 at 16-degrees angle of attack.

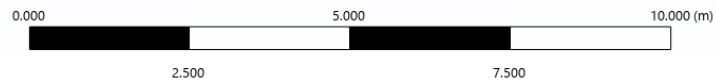
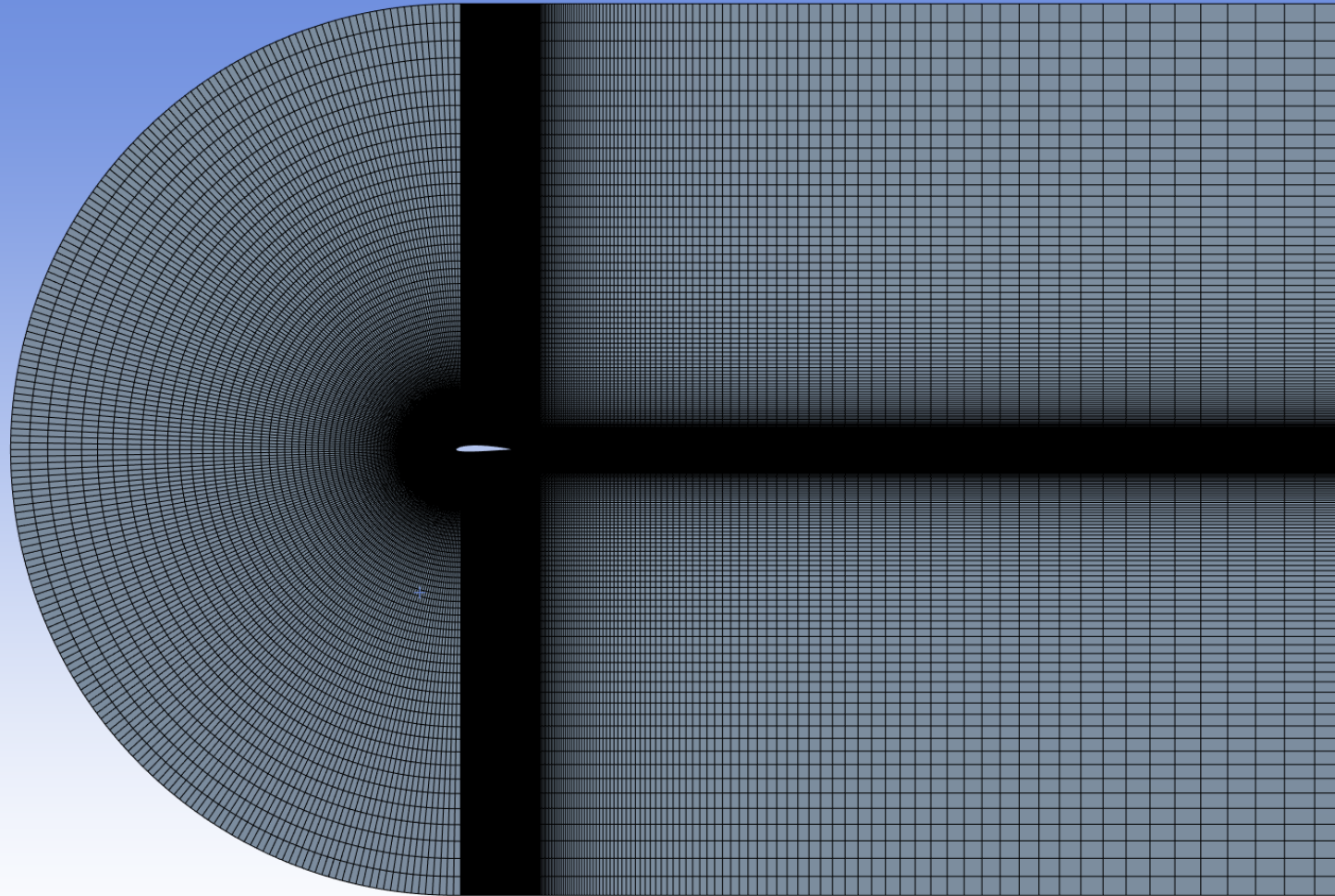


Figure 3.
Full
Mesh
2412 Airfoil

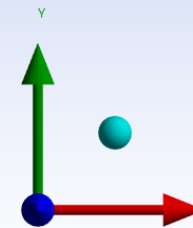
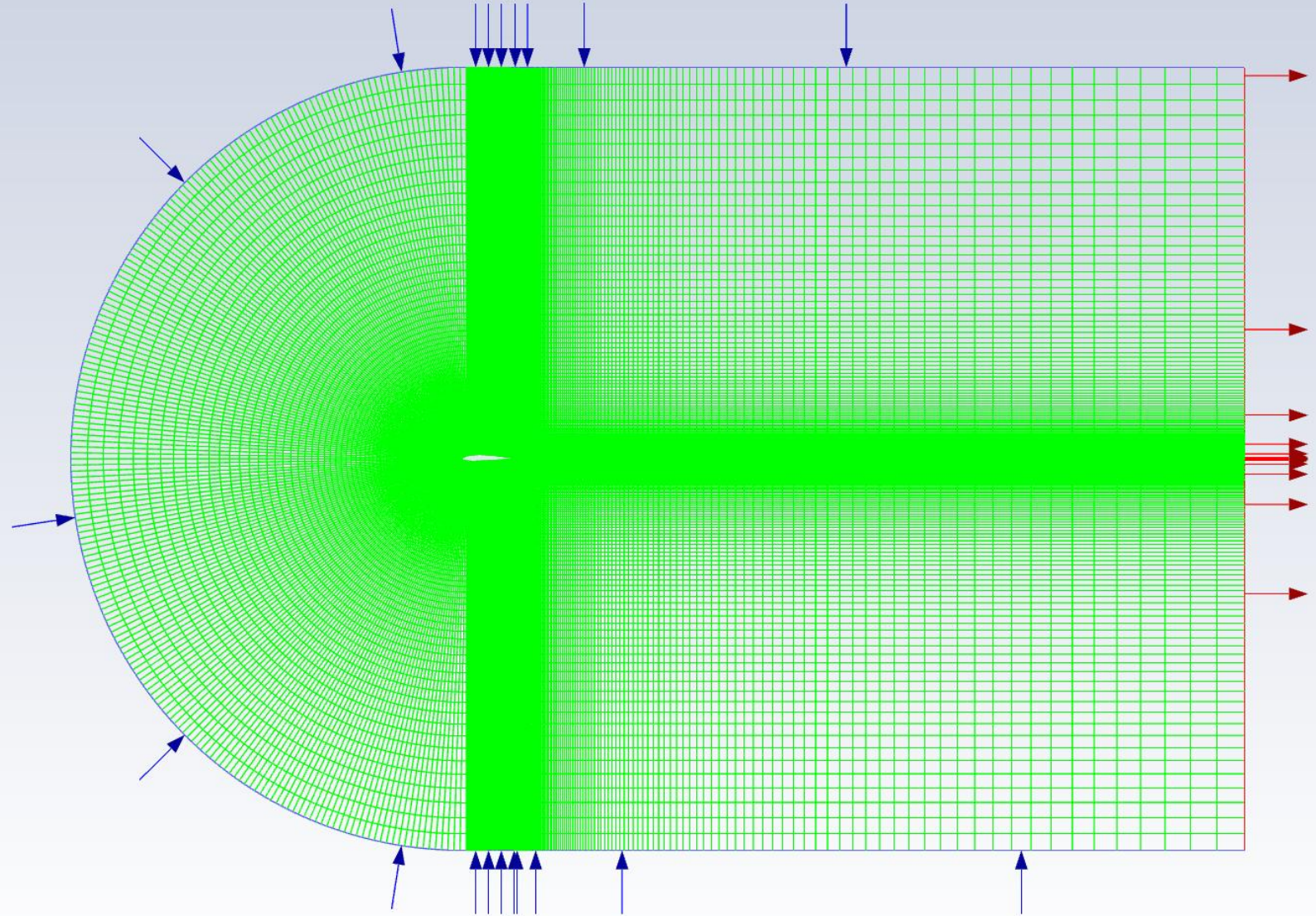
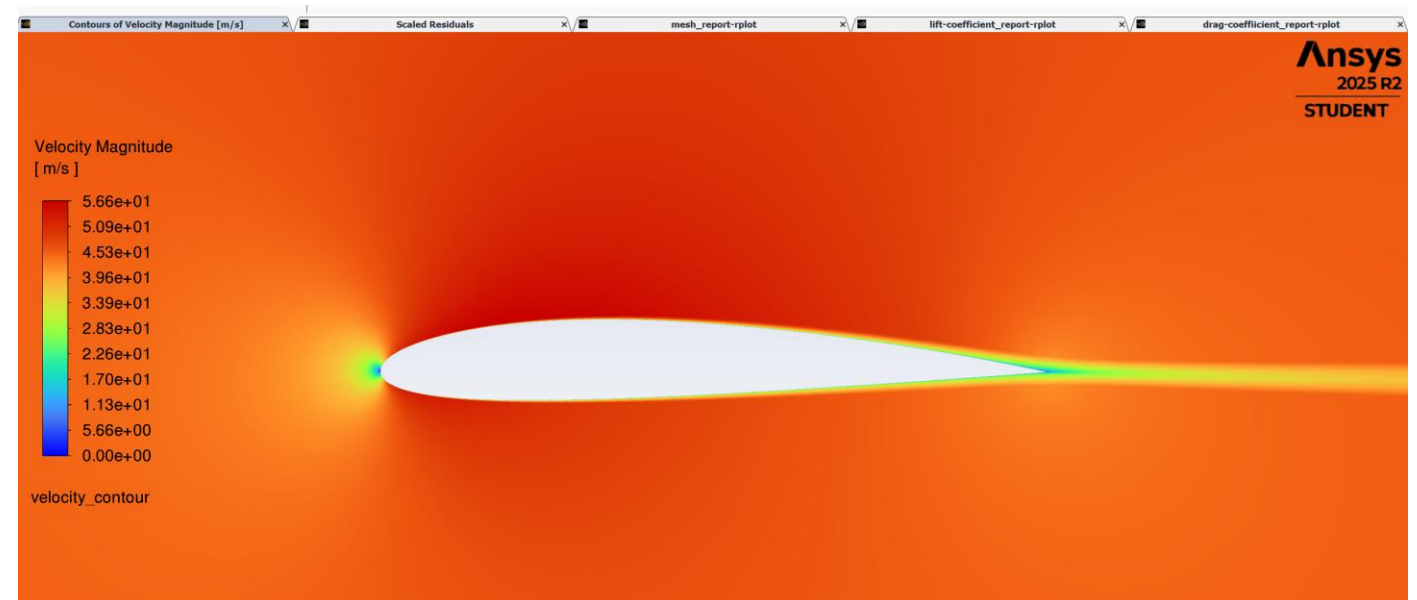
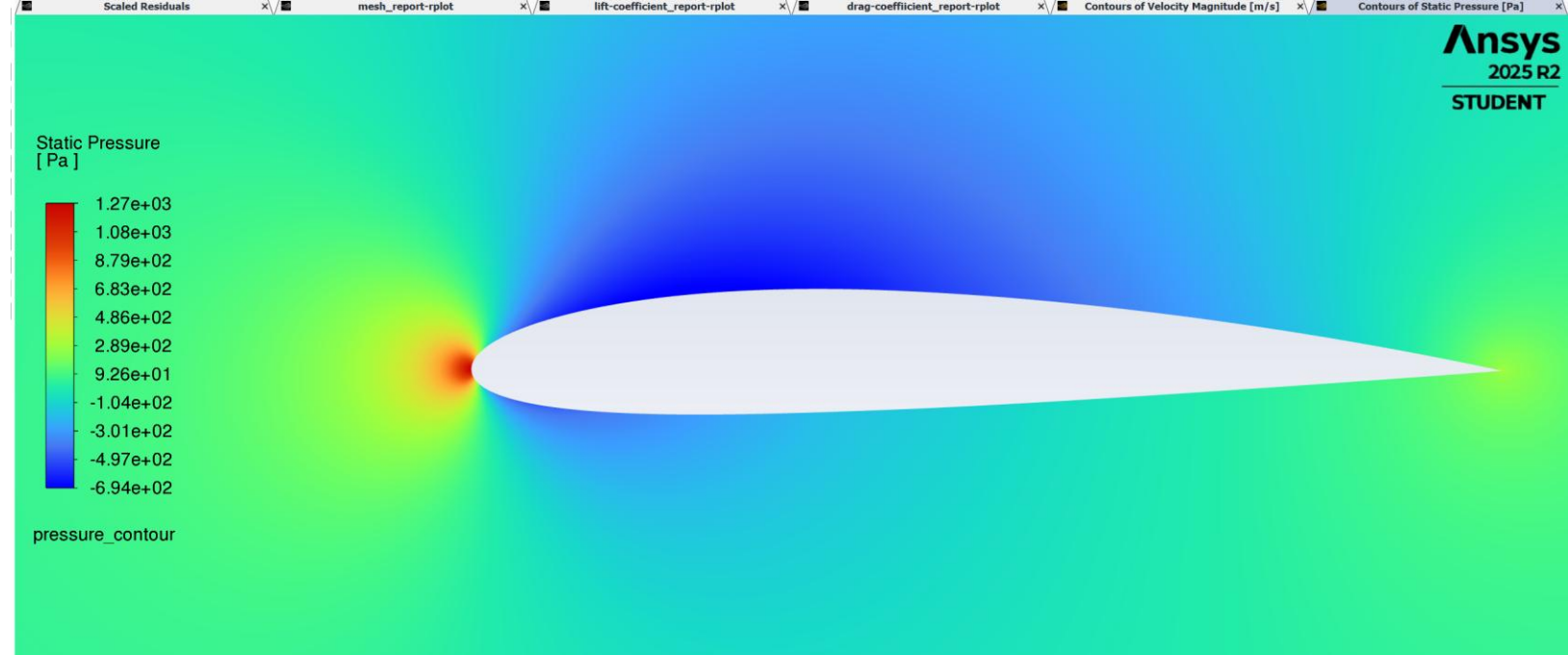


Figure 4.
Inlet and
Outlet in
Fluent



Figures 5 and 6

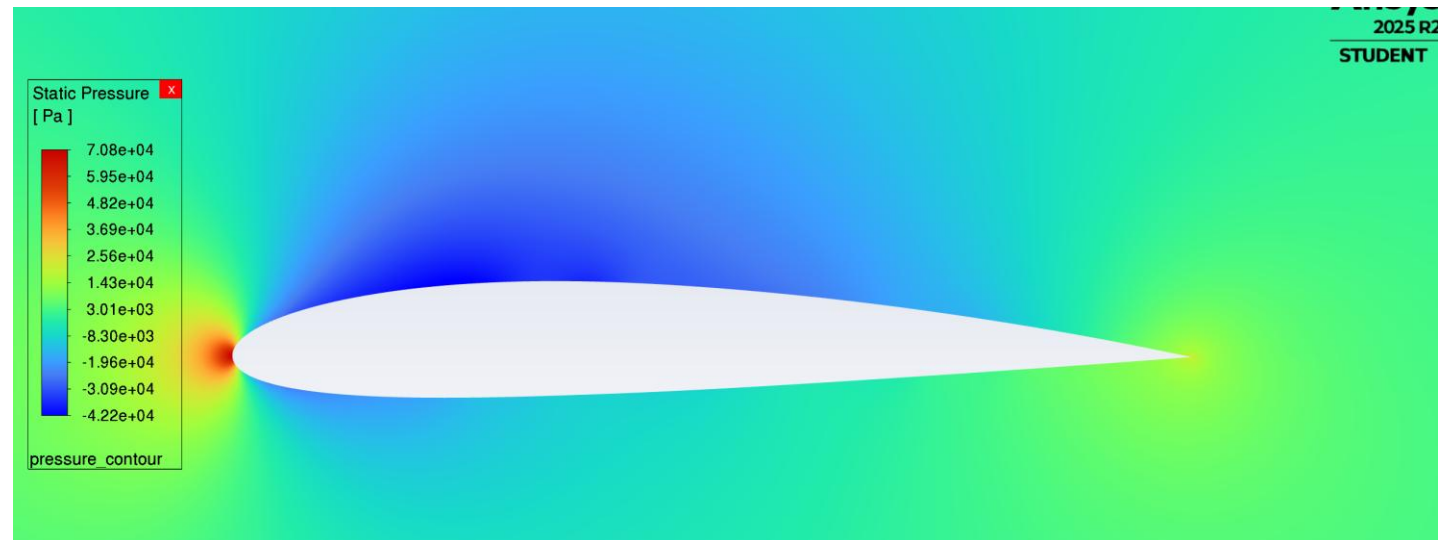
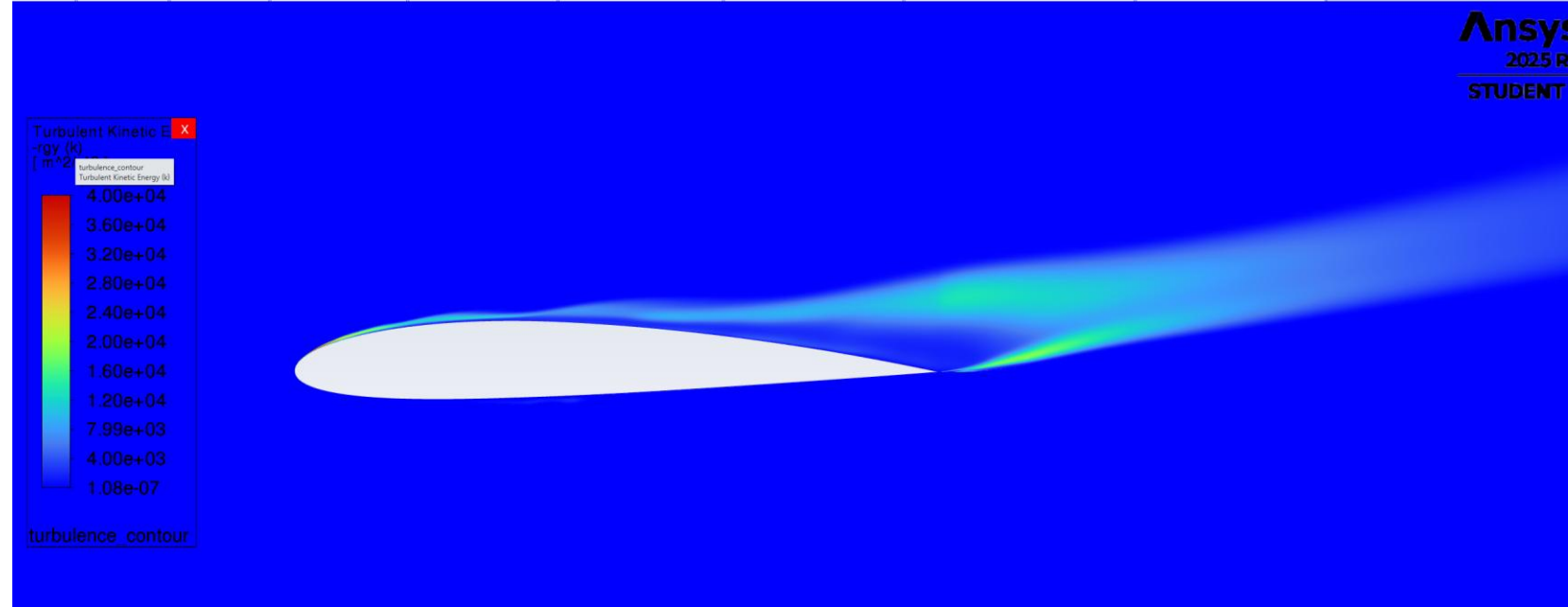
Pressure and
Velocity Contours at
Mach 0.13 (45 m/s)
at 0-degree angle.



Figures 7 and 8

Pressure and Turbulence Contours at Mach 1.5 (497 m/s) at 16-degree angle.

- A pressure-based solver would have been a better choice to get the shock waves instead of a density-based solver setting as it simpler to set up and avoid sensitivity issues.
- A turbulent wake is seen in Figure 4.



Validation and Verification

- Compared Fluent results to NACA 2412 experimental data.
- Simulated Lift data was accurate while drag coefficient was overestimated.

Table 1. Fluent vs NACA 2412 Results at Mach 0.13 (45 m/s)

Angle of Attack	Fluent Lift Coefficient	Fluent Drag Coefficient	NACA Lift Coefficient	NACA Drag Coefficient	Percent Difference
0	2.15 e-01	9.27 e-03	2.5 e-01	6.5 e-03	15.1% and 35.1%
8	1.049	1.42 e-02	1.05	1.15 e-02	0.0095% and 21.0%
16	1.636	4.265 e-02	1.50	2.30 e-02	8.67 and 59.9%



Key Results

- **Validation:** Validation at Mach 0.13: Lift coefficients closely matched NASA data; drag less accurate due to mesh resolution.
- **Results of Compressible flow (Mach 1.0 and 1.5):** Density solver captured turbulent wake; signs of shock interactions but not fully demonstrated at Mach 1.5, 16° angle of attack.
- **Outcome:** Demonstrated Fluent's capability of modeling subsonic and compressible flows over a simple airfoil geometry.

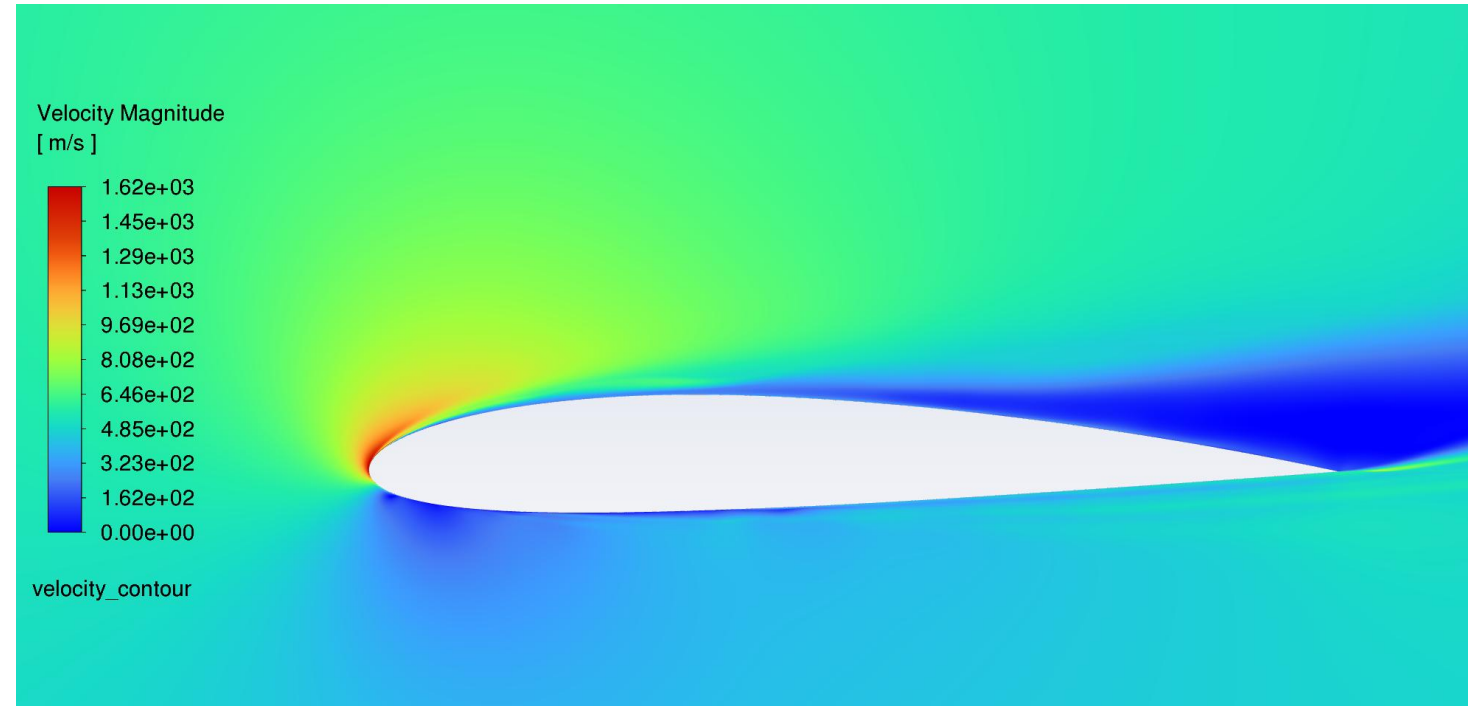
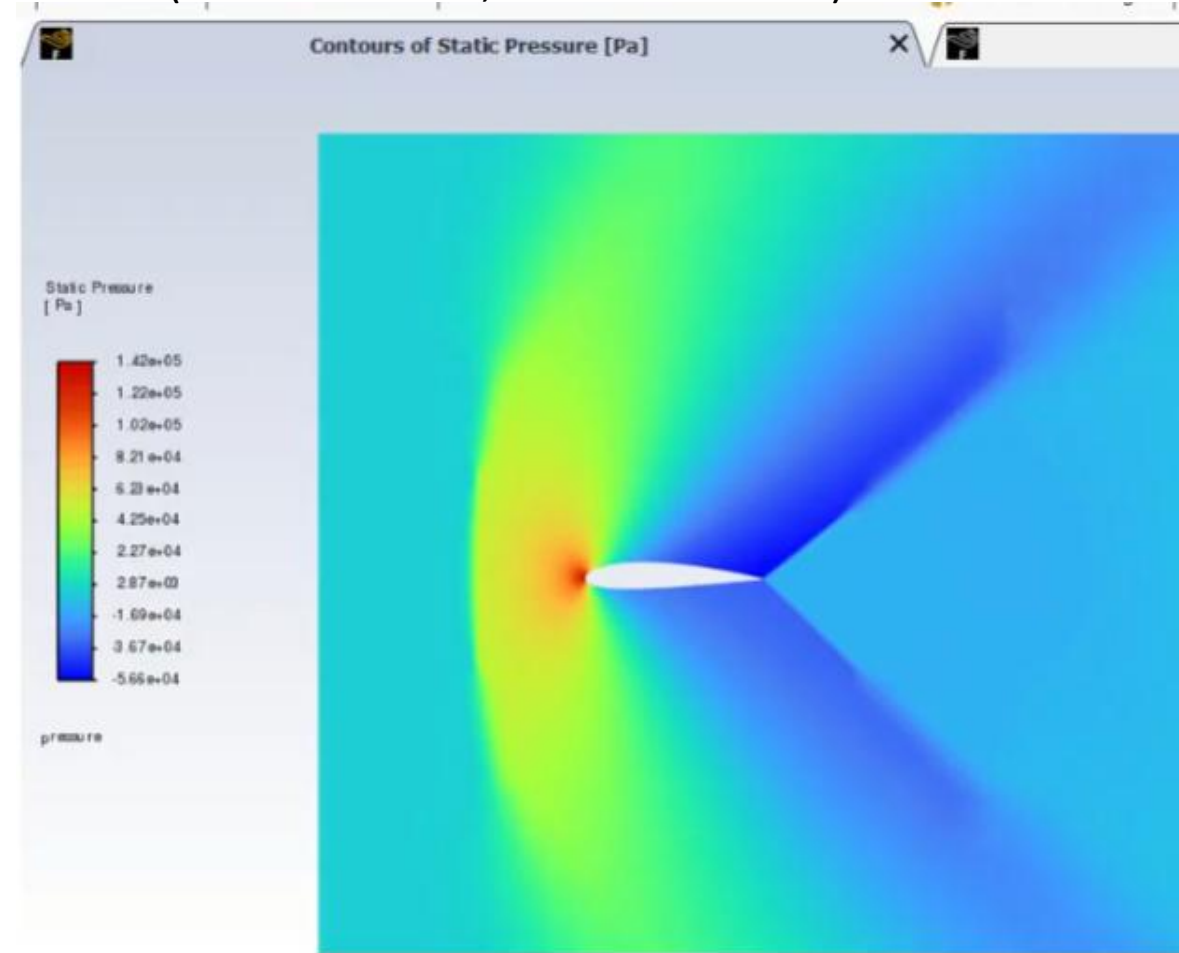


Figure 9. Velocity contour at Mach 1.5 (497 m/s) at 16° Angle of Attack.

Conclusion

- **Challenges:** Mesh generation (bias reversal) and solver settings for turbulence modeling
- **Limitations:** Drag prediction limited by steady-flow assumptions and mesh resolution.
- **Improvement:** redo Mach 1.5 simulation with pressure-based and density-based solver to resolve shock waves.
- **Future work:** unsteady simulations, 3D airfoils, and more complex geometries.

Figure 10. Pressure Contour of Flow over 2D Airfoil at Mach 1.2 Using Pressure-Based Solver (Armin Hashemi, YouTube Tutorial)



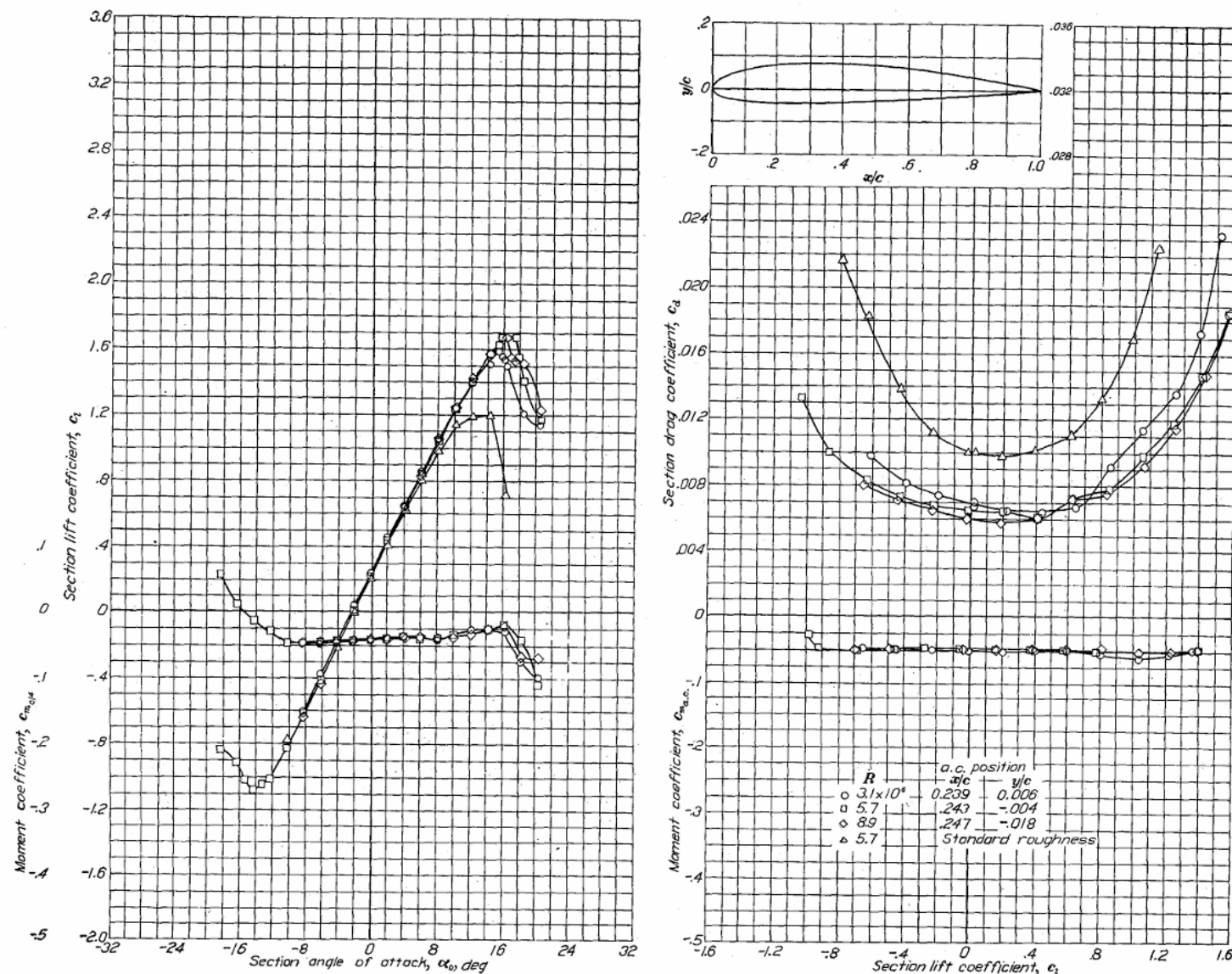
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- Cillian Thomas. NACA2412 Tutorial in ANSYS Fluent (Student Version). (2023). YouTube. <https://www.youtube.com/watch?v=3i9Ryq-m1HA>.
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- Kundu, P. K., Cohen, I. M., & Dowling, D. R. (2015). Fluid Mechanics (6th edition). Academic Press.
- Sandoval, G. (2025). Incompressible Flow Over a 2D NACA 2412 Airfoil. Michigan Technological University, ME 5210 CFD Project Report.



Appendix A

- **Figure 11.** is the NACA 2412 Experimental Data at Mach 0.13.



Appendix B

- **Figure 12.** is the Streamlines of the 2412 at Mach 0.13.

